WISE Seminar

GNN-IP Inference Optimization Reconstruction

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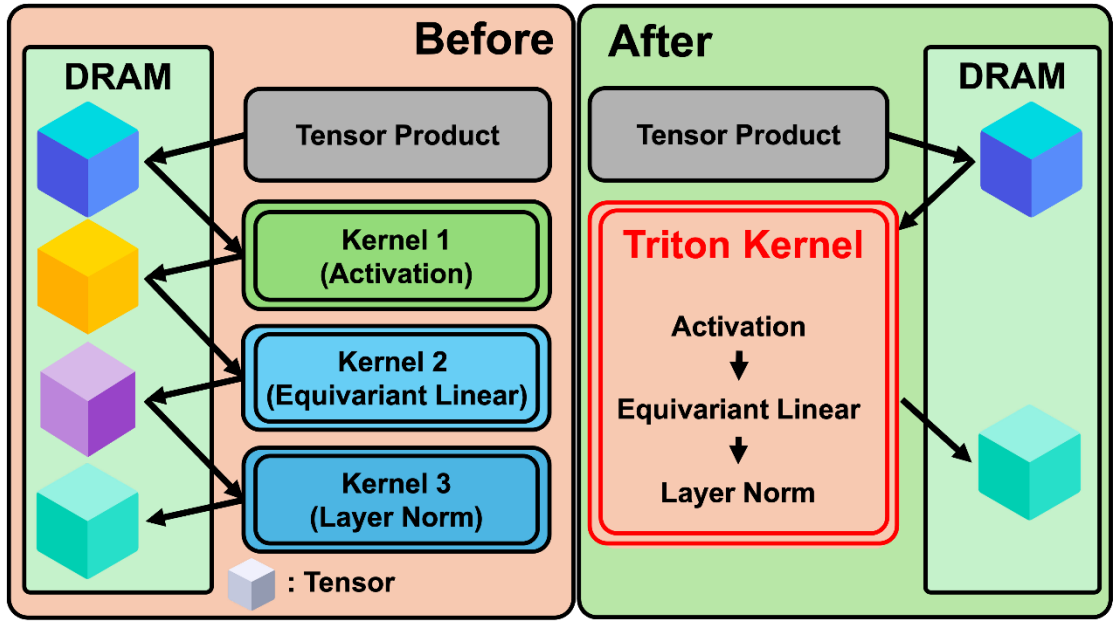
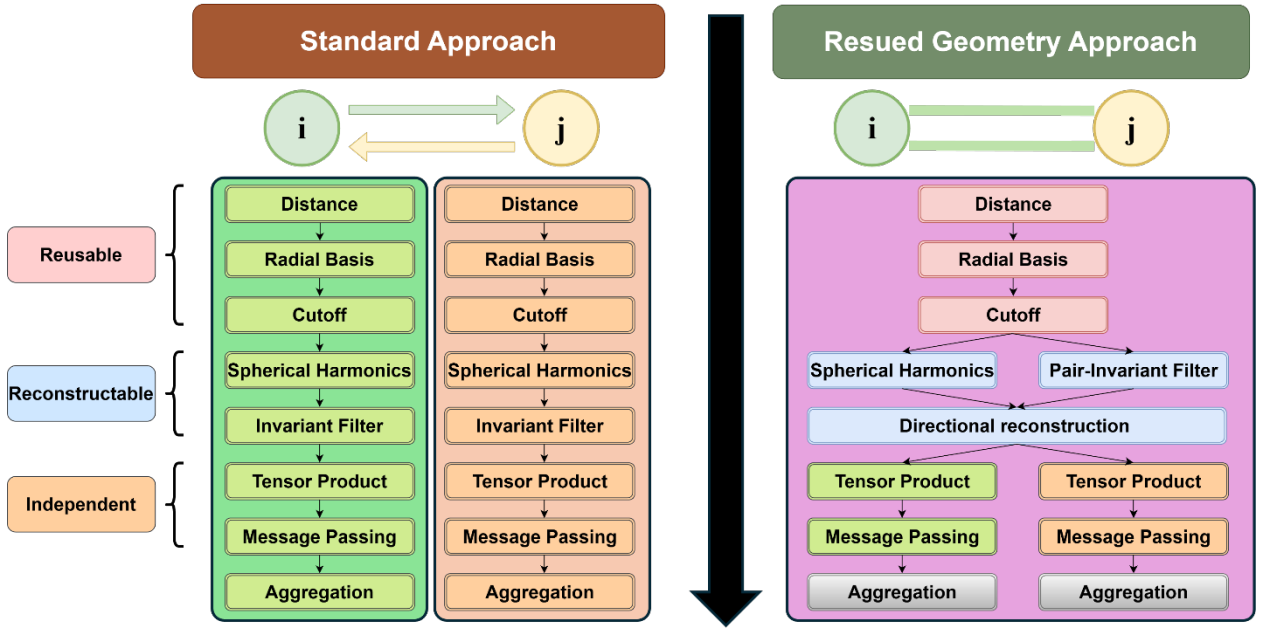
GNN-IP R&D Proposal

3 : Optimization of GNN-IP Kernels and Graph Computation





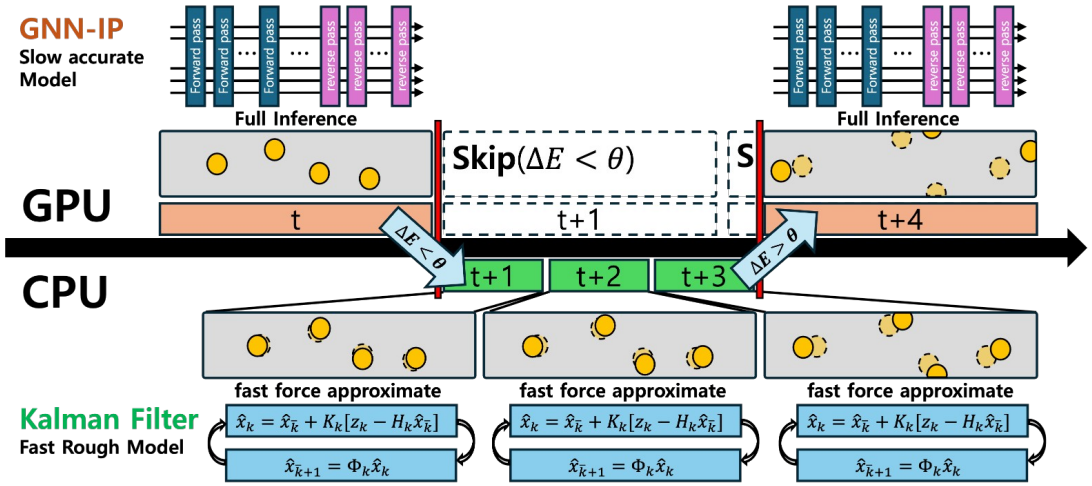
Kernel optimization for non–tensor product (Non-TP) operations

Elimination of redundant computations through restructuring of input representation

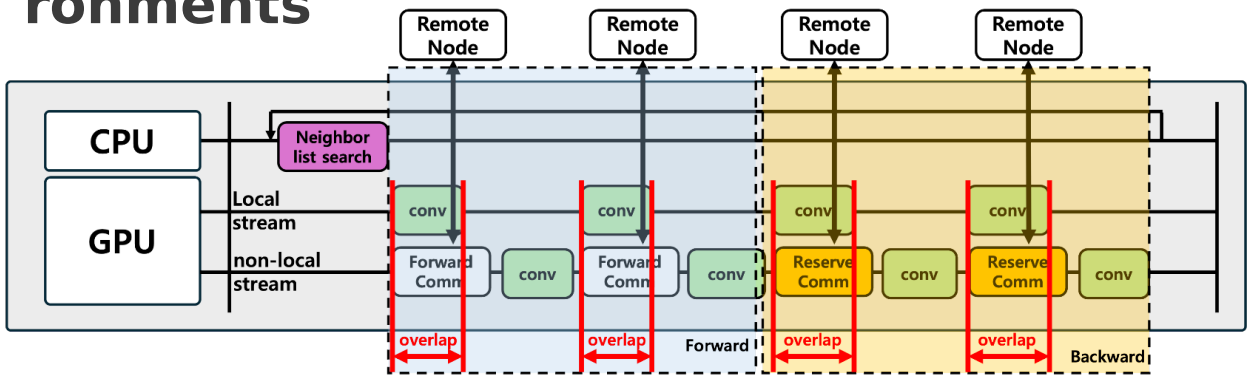
GNN-IP kernel level optimization

GNN-IP R&D Proposal

Year 4 : Optimization of GNN-IP Kernels and Graph Computation



Year 5: Acceleration of GNN-IP Inference in Multi-GPU / Multi-node Environments

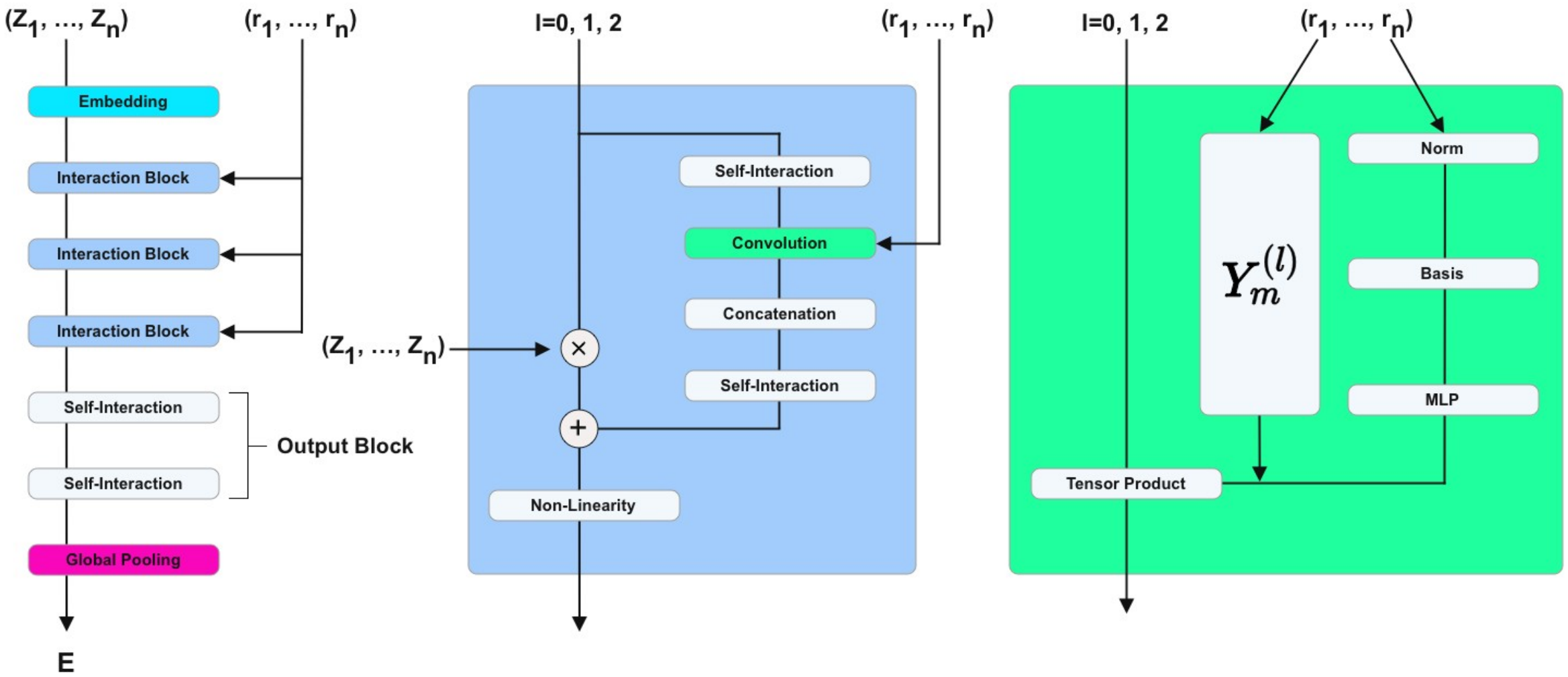


- **Design of a training-free adaptive skipping policy leveraging temporal redundancy**

Scalable GNN-IP inference acceleration for multi-GPU and multi-node simulation settings

GNN-IP runtime Optimiza ***GNN-IP Distributed System Optimiza***

Equivariant Model PipeLine (NequIP)



- NequIP: Batzner, S. et al., E(3)-equivariant graph neural networks for data-efficient and accurate interatomic potentials. Nat Commun 13, 2453 (2022).

Problem

- **Pair-invariant**
  - **Distance**
  - **Radial basis**
  - **Cutoff**
- **Direction-dependent**
  - **Spherical harmonics**
  - **Weighted filter (NequIP)**
- **Non-reusable**
  - **Message**
  - **Tensor product**
- **Reconstruct the data and computation flow while preserving the original GNN-IP formulation and learned parameters.**

Challenge

1. **Structural Inefficiency: Edge-wise Redundancy**
   - **Redundant Geometric Computation**
     - Standard NequIP processes each directed edge (i to j and j to i) independently, leading to duplicated calculations for distance and radial basis
   - **Pair-wise Reconsturing**
     - Input flow must be restructured to Pair-based to enable "Compute once, reuse via sign flip"

Challenge

2. Integration of FlashTP & JIT Compilation

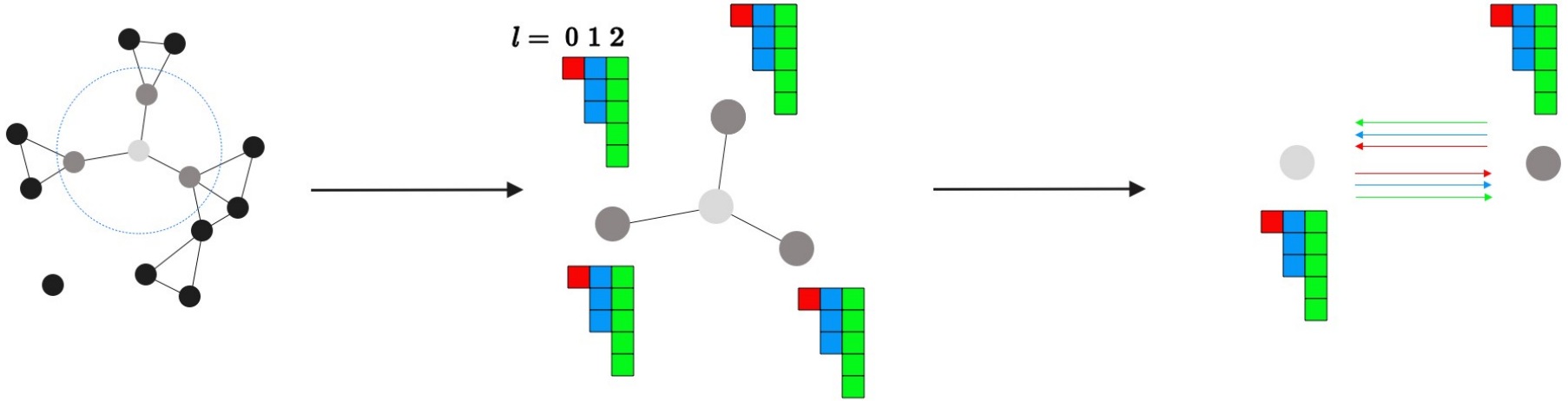
- **Synergy with FlashTP**
  - Beyond simple TP acceleration, the method requires tight integration of FlashTP-style fused kernels with pair-wise reuse
- **High Implementation Difficulty**
  - JIT (Just-In-Time) Compilation: Generating and compiling graph-specific CUDA/Triton code at runtime
  - Dynamic Topology: Significant challenge in building optimized kernels tailored to varying graph structures during inference

Related Work

- **FlashTP, SevenNet : System-level Acceleration**
  - **Focuses heavily on Tensor Product (TP) fusion and parallel scaling, yet leaves the upstream bidirectional geometric redundancies (e.g., Spherical Harmonics computation for i to j and j to i) completely unaddressed.**
- **NEMP : Algorithmic Modification**
  - **Reduces overhead by aggregating features in the node space, but fundamentally alters the original E(3)-equivariant message-passing formulation and its expressivity.**
- **I've not yet found any prior work that removes this rebundacy while strictly preserving the original formulation**

- NequIP: Batzner, S. et al., E(3)-equivariant graph neural networks for data-efficient and accurate interatomic potentials. Nat Commun 13, 2453 (2022).
- SevenNet: MDIL-SNU., SevenNet - a graph neural network interatomic potential package supporting efficient multi-GPU parallel molecular dynamics simulations. J. Chem. Theory Comput. (2024).
- NEMP: Zhang, Y., & Guo, H., Node-Equivariant Message Passing for Efficient and Accurate Machine Learning Interatomic Potentials. Chem. Sci. (2025).
- FlashTP: Lee, et al., FlashTP: Fused, Sparsity-Aware Tensor Product for Machine Learning Interatomic Potentials. ICML (2024).

Proposed Mehtod



- **E(3)-equivariant models' data expression be like Rotation + Translation + Reflection**
  - **Rotation : Spherical Harmonics : SO(3) irreps expression (scalar + vector + tensor)**
  - **Translation : Absolute positions are removed by using relative coordinates so model become invariant**
  - **Reflection : Even / Odd parity is determined by the degree l**
  - **Reflection symmetry is handled via parity transformation**

- Batzner, S., Musaelian, A., Sun, L. et al. E(3)-equivariant graph neural networks for data-efficient and accurate interatomic potentials. Nat Commun 13, 2453 (2022).

Reason Why choose this

- 1. The Bottleneck: Spherical Harmonics
  - Even with modern Cartesian-polynomial implementations, SH remains a visible geometric cost when the graph is large.
  - Profiling Insight: In our lab baseline profiling, SH takes roughly 18-26% of model time on large dense cases such as MPtrj and MD22 nanotube.
- 2. Mathematical Foundation: Parity Property
  - The inputs for bidirectional edges (i to j and j to i) contain identical geometric information, differing only by an equivariant transformation.
- 3. Proposed Solution: Sign Flip Reuse
  - Compute Once: Calculate spherical harmonics only for one canonical direction within each pair.
  - Instant Reuse: Recover the reverse edge by parity sign, instead of recomputing the same geometry.

Proposed Method

air Metadata Construction

(i to j) and (j to i) into one undirected pair and keep forward / reverse indices explicitly.

air-shared Geometry

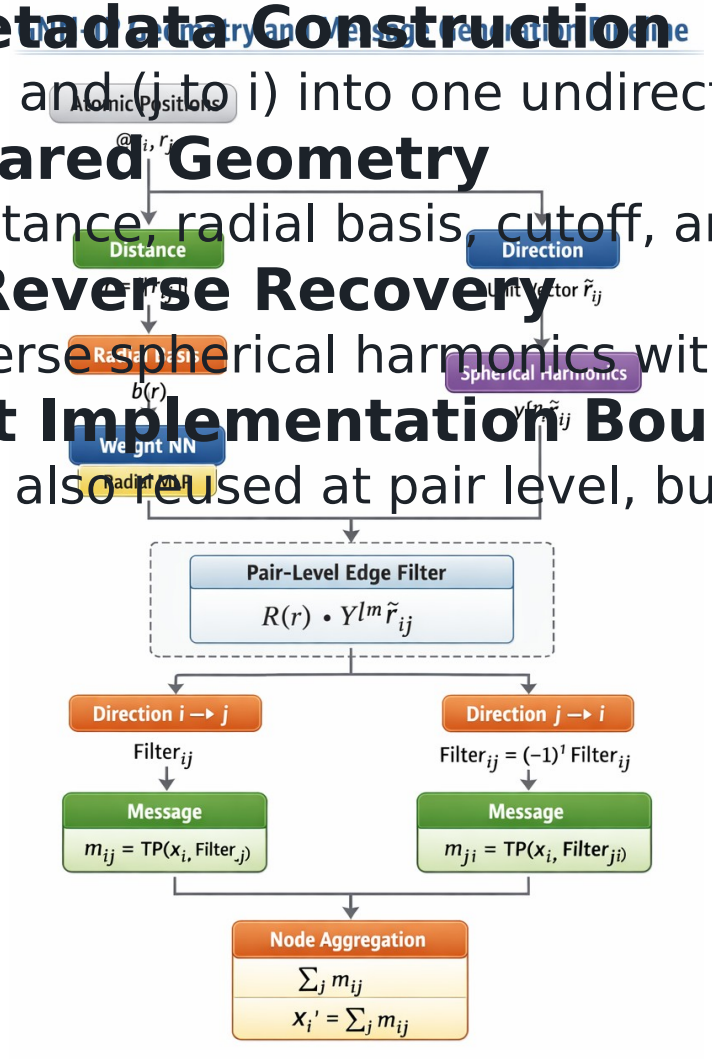
ute distance, radial basis, cutoff, and spherical harmonics once per pair instead of once per directed

xact Reverse Recovery

er reverse spherical harmonics with parity sign, not with a second SH evaluation.

urrent Implementation Boundary

t_nn is also reused at pair level, but tensor product and aggregation are still executed on directed ed



- **Message composition:**
- **Message reverse:**
- **Spherical Harmonics parity:**
- **The inputs contain the same information; they diffe only by an equivariant transformation.**

Enables instant reuse of pre-computed values throug a sign flip.

Experiment Plan

- Baselines
  - SevenNet e3nn baseline / SevenNet + pair execution(full) / FlashTP reference
- Workloads
  - Small sparse: SPICE 2023, ANI-1x
  - Small dense: phononDB, selected periodic crystals
  - Large sparse: OMol25 validation, OC20 validation
    - Large dense: MPtrj, MD22 nanotube, OMat24
  - Evaluation Metrics
    - Energy-force exactness
    - Cold / steady latency + stage profiling + GPU utilization
    - Size-density analysis and quadrant comparison
  - Scope: single-GPU lab benchmark first, multi-GPU scaling next

Discussion

- Modern SH Implementation
  - Uses fast Cartesian polynomials, so the win is not from replacing slow trigonometric code.
- Current Bottleneck
  - The real gain comes from removing duplicated geometry, SH, and pair-weight work on reverse edges.
- Current Limitation
  - TP and aggregation are still directed-edge operations, so the benefit is strongest only on large dense graphs.
- Next Step
  - Pair-major TP plus backend co-design is required before claiming consistent end-to-end acceleration over FlashTP.